Unraveling the polarimetric difference of Repeating and Non-repeating FRBs

Chengwei Liang (GZHU)

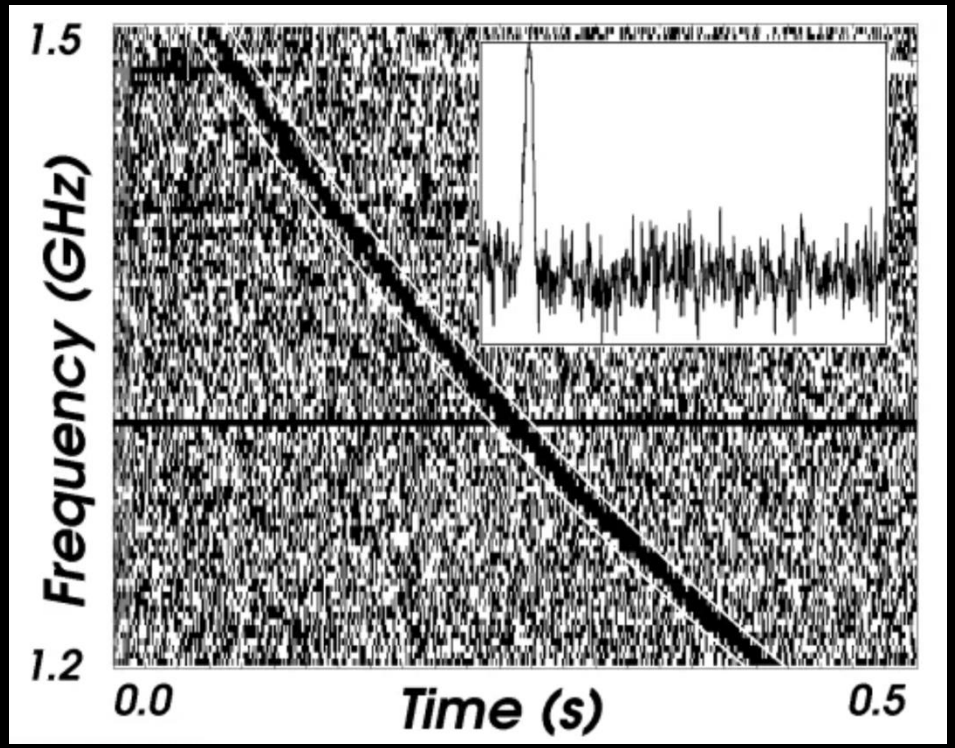
Supervisor: Rui Luo (GZHU)

Shiqian Zhao (GZHU), Yao Chen (GZHU), Xiaohui Liu (NAOC), Jingwen Chen (GZHU)

Australasian Pulsar & FRB Workshop.
May. 2026

Background

- First discovered in 2007 (Lorimer burst)
- Bright millisecond-duration radio transients
- Extragalactic origin (large dispersion measures)
- First repeating source: FRB 20121102

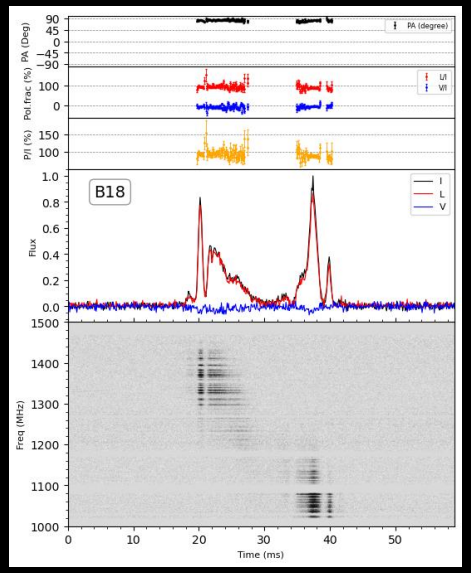


D. R. Lorimer et al. 2007

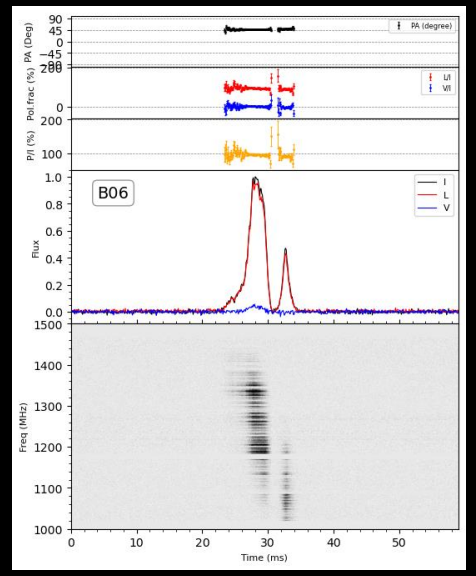
Key Properties of Repeating FRBs

- Repeating FRBs show rich temporal–spectral structures
- Downward frequency drifting sub-bursts
- Narrow-band emission features
- Suggest structured emission process

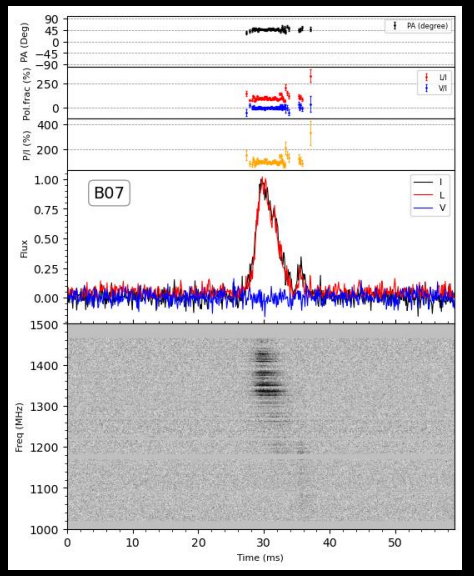
→ Indicate possible physical differences from one-off bursts



Muticomponent



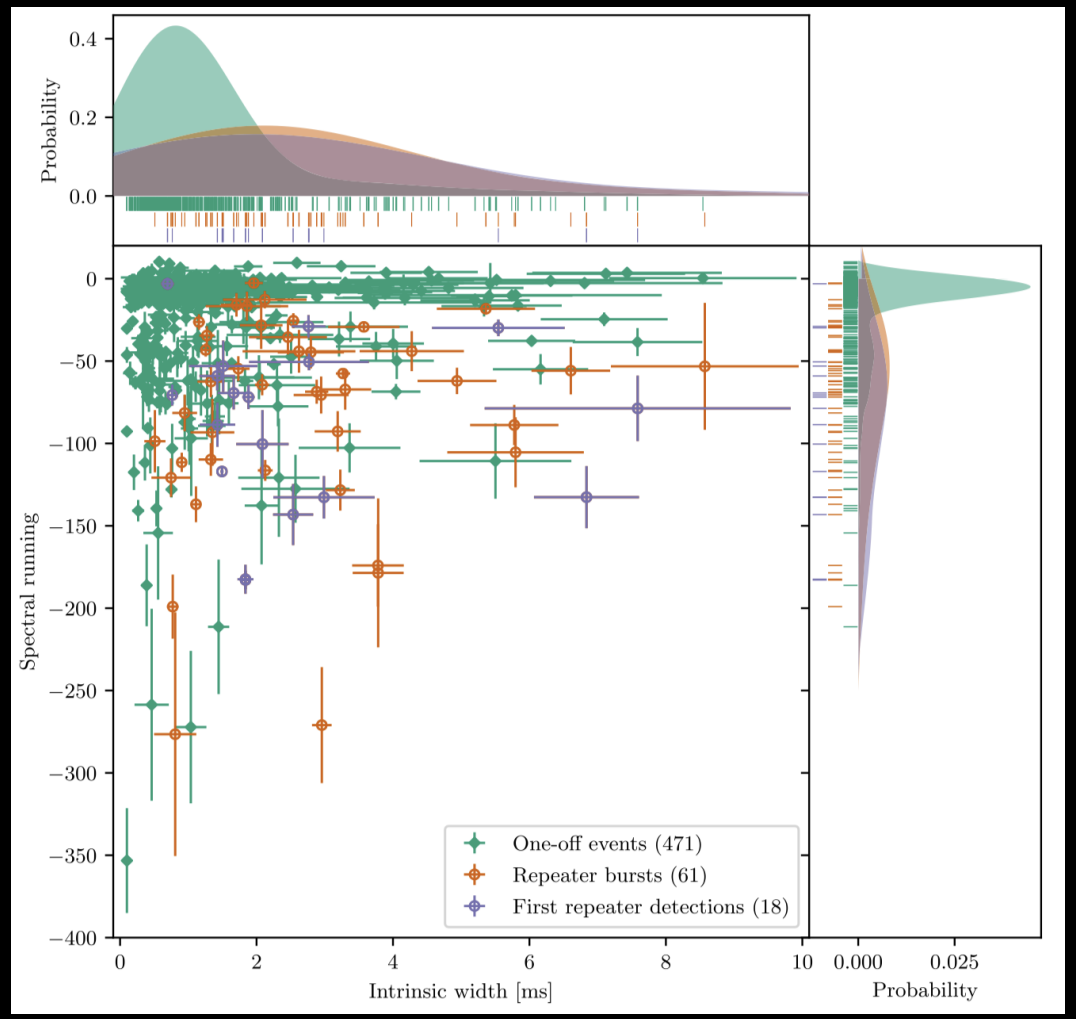
Drift



Narrow-band emission

Background

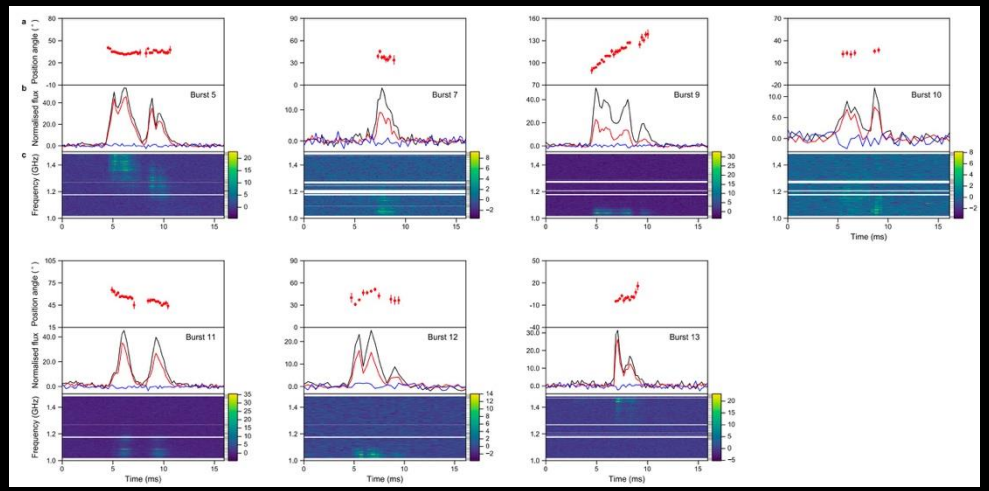
- 4020 FRB sources detected, including 101 repeating sources
- FRBs can be divided into repeating and apparently non-repeating sources
- Statistical studies show differences in spectral properties



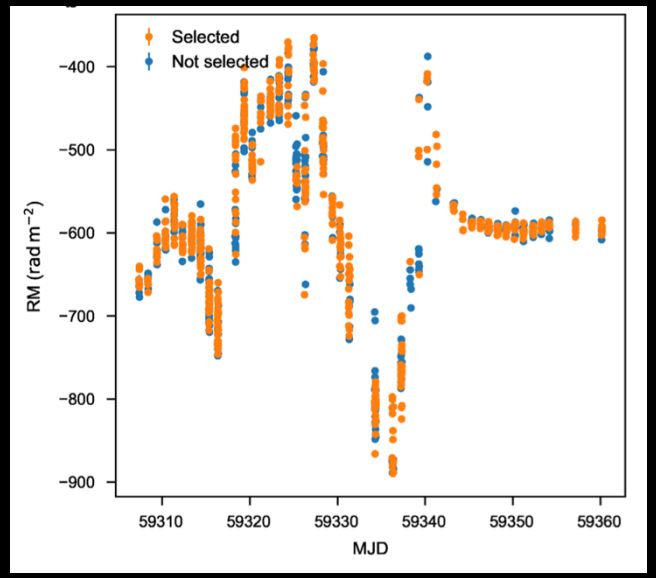
Pleunis et al. 2021

Background

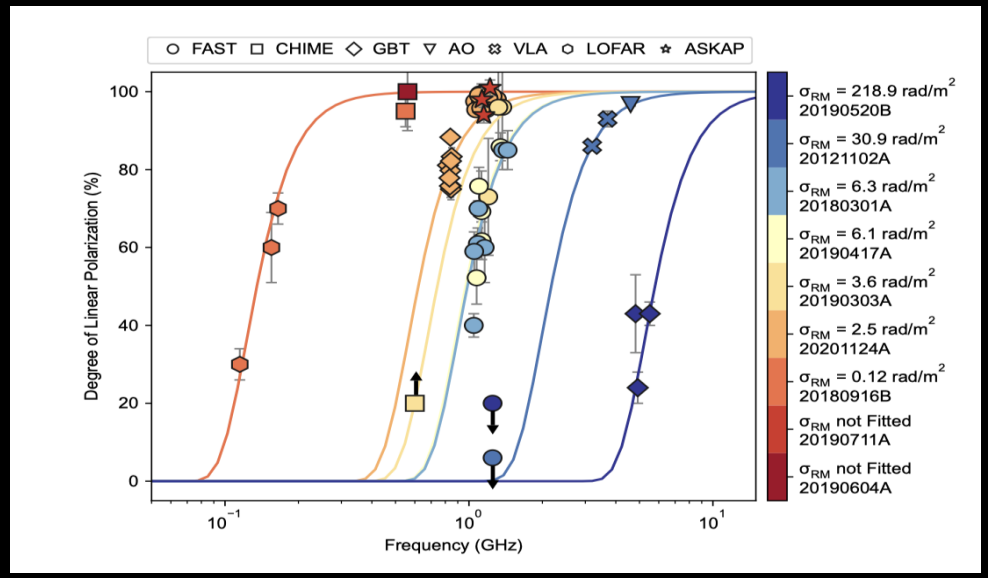
- Polarization probes emission mechanisms and magnetic environments



Luo et al. 2020



Xu et al. 2022

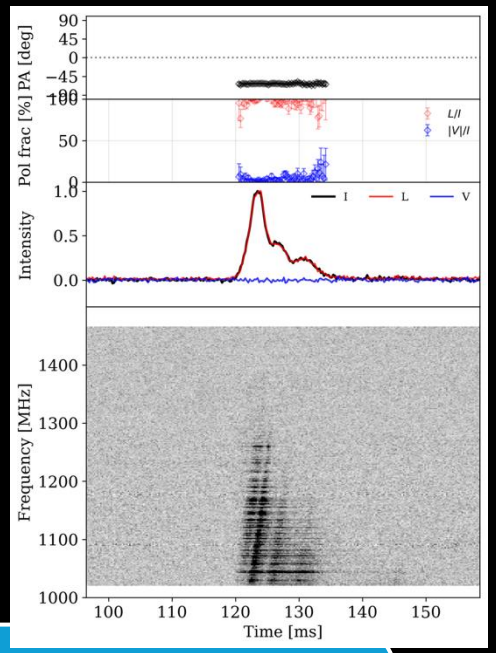


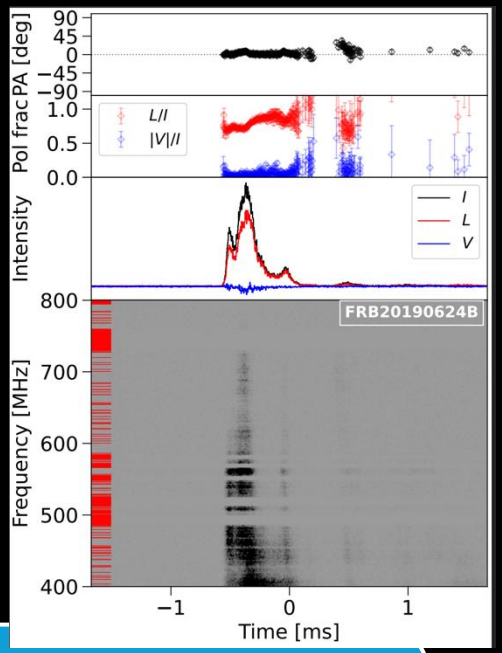
Feng et al. 2022

Motivation

- Statistical differences exist but are not conclusive
- Polarization probes physical environments
- Lack of quantitative comparison between populations
- Is there a fundamental difference between repeating and non-repeating FRBs?

Data Set





Repeating FRBs:

536 bursts from FRB 20201124A observed by FAST

Jiang et al. 2024

Non-repeating FRBs:

63 bursts from the first CHIME/FRB Baseband catalog

Pandhi et al. 2024

Features

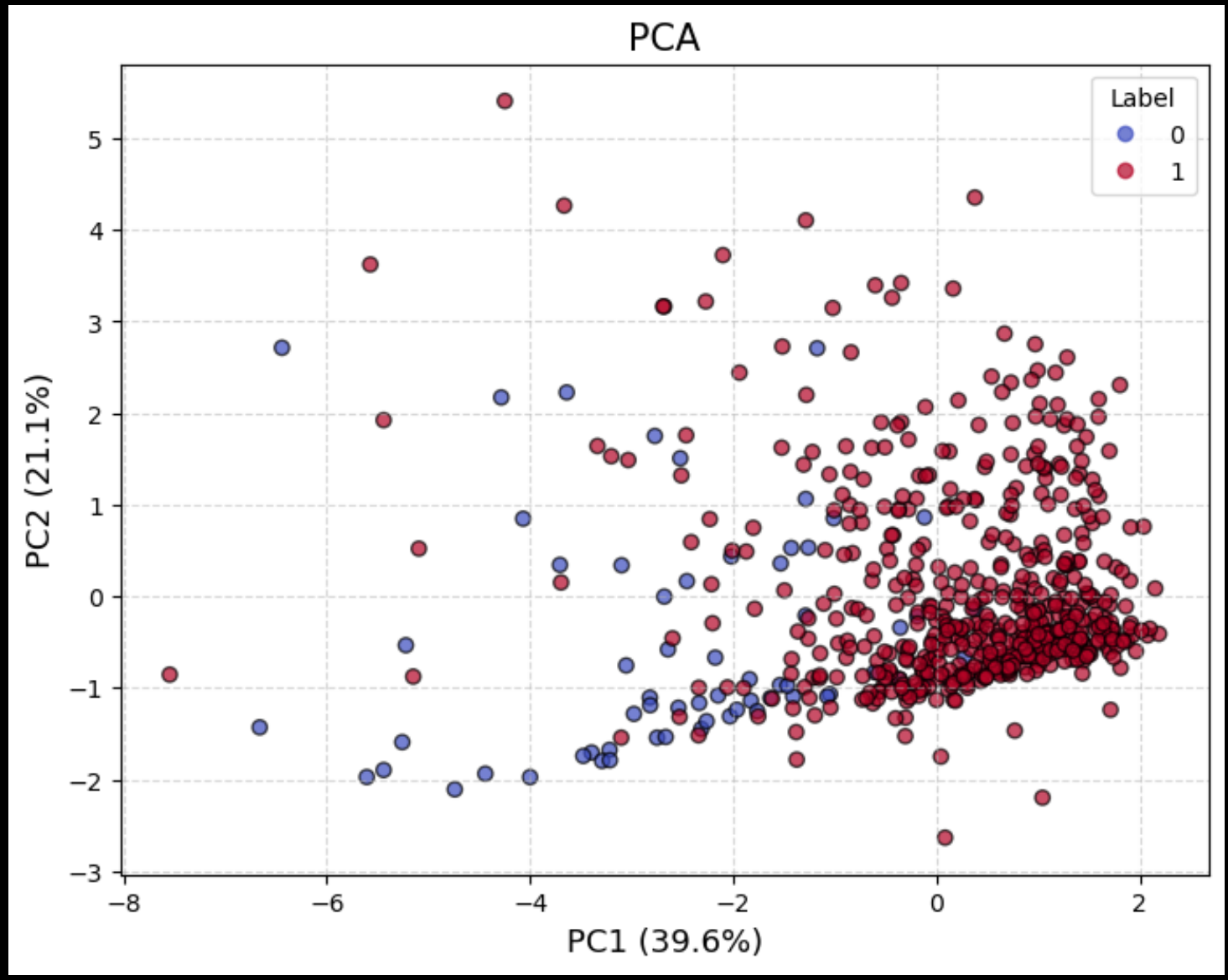
Features	Interpretation
Mean	Average PA over the burst duration
Standard	Dispersion of PA around the mean
Avg_change	Average absolute change between continuous measurements
Autocorr1	Lag-1 autocorrelation
Trend_slope	Slope of linear fit to PA vs time
Trend_r2	R-squared of linear trend

Dimensionality Reduction Methods

PCA (Principal Component Analysis)

PC1 + PC2 explain 60% of total variance

Result: No visible class separation

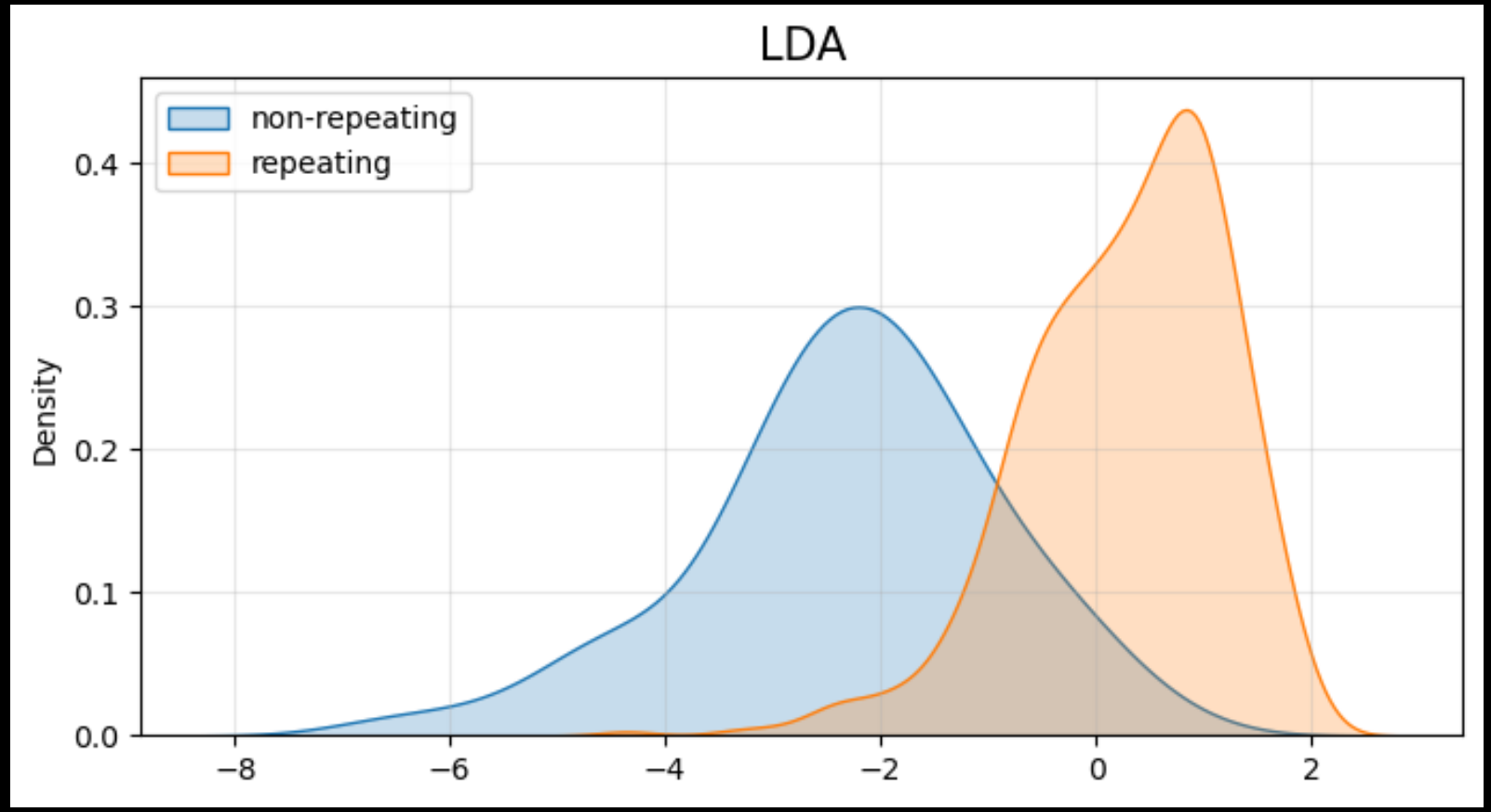


Dimensionality Reduction Methods

LDA (Linear Discriminant Analysis)

Maximize the separation between classes

Result: Partial separation



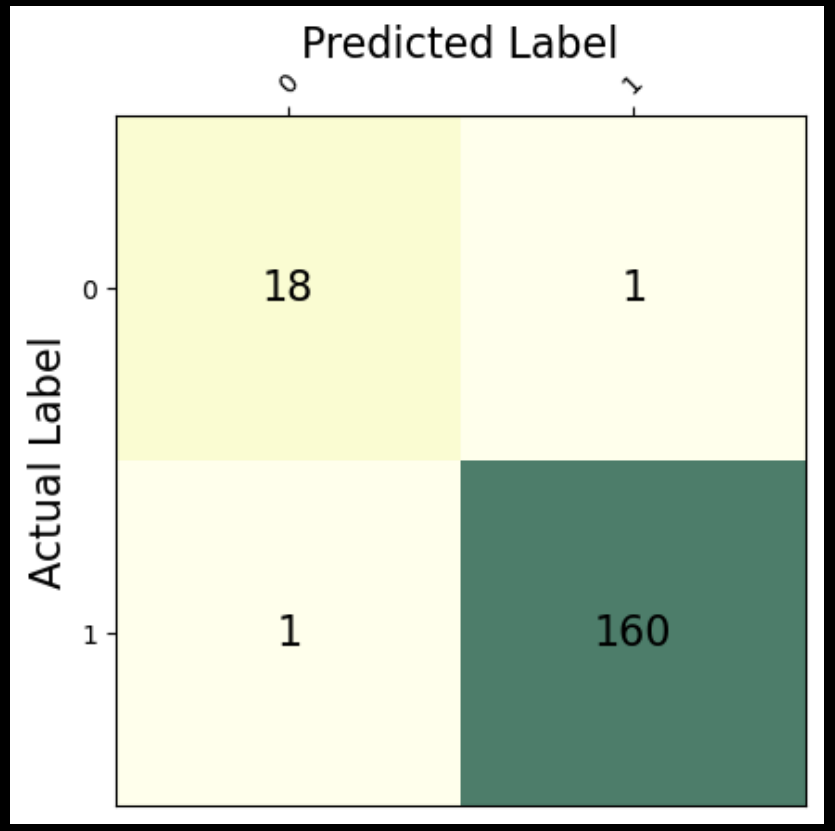
Classification Methods - XGBoost

	Original train-set	Processed train-set
Non-repeating FRB (0)	44	374
Repeating FRB (1)	375	374

Smote + Tomek

	Precision	Recall	F1-score
Non-repeating FRB (0)	0.95	0.95	0.95
Repeating FRB (1)	0.99	0.99	0.99

Classification report



Discussion

Interpretation
- Statistical separability in polarization feature space
- PA variation plays a key role
- Differences are not captured by variance (PCA), but by class structure (LDA)

Physical Implications
- Polarization reflects emission geometry and environment
- Results are consistent with differences in magneto-ionic environments

Methodological Considerations
- Supervised vs unsupervised methods reveal different structures
- Nonlinear classifier captures hidden separations

Future Directions
- Larger and more uniform samples
- More physically motivated features
- Toward time-dependent / structured polarization analysis

THANKS!